

# Myungje\_Kim\_Final\_Project

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```
library(tidyverse)
library(openintro)
library(gridExtra)
```

```
data("birds")
```

This data has phase of flight and time of day as factors. birds adj 2 and 3 is just to show that 16671 has the same observations when I filter it with NAs one by one.

```
birds_all_states <- birds |>
  select(phase_of_flt, time_of_day, state) |>
  na.omit()

birds_CA <- birds |>
  select(phase_of_flt, time_of_day, state) |>
  na.omit() |>
  filter(state == "CA") |>
  mutate(
    phase_of_flt = as.factor(phase_of_flt),
    time_of_day = as.factor(time_of_day)
  )

birds_CA
```

```
# A tibble: 1,540 x 3
  phase_of_flt time_of_day state
  <fct>         <fct>    <fct>
1 Approach     Day      CA
2 Landing Roll Day      CA
3 Take-off run Day      CA
4 Approach     Day      CA
```

```

5 Approach      Day      CA
6 Approach      Day      CA
7 Climb         Day      CA
8 Landing Roll Day      CA
9 Take-off run Day      CA
10 Approach     Night    CA
# i 1,530 more rows

```

```

birds_Otherst <- birds |>
  select(phase_of_flt, time_of_day, state) |>
  na.omit() |>
  filter(state != "CA") |>
  mutate(
    phase_of_flt = as.factor(phase_of_flt),
    time_of_day = as.factor(time_of_day)
  )

nrow(birds_CA) + nrow(birds_Otherst) == nrow(birds_all_states)

```

```
[1] TRUE
```

```

#birds_adj2 <- birds |>
# filter(!is.na(phase_of_flt)) |>
# mutate(
#   phase_of_flt = as.factor(phase_of_flt)
# )

#birds_adj3 <- birds_adj2 |>
# filter(!is.na(time_of_day)) |>
# mutate(
#   time_of_day = as.factor(time_of_day)
# )

#birds_adj2
#unique(birds_adj2$phase_of_flt)

#birds_adj3
#unique(birds_adj2$time_of_day)

```

Checking if there is NA values in each dataset.

```
#birds |>
#  select(state) |>
#  count(state == "NA")

#birds_CA |>
#  select(state) |>
#  count(state == "NA")

#birds_Otherst |>
#  select(state) |>
#  count(state == "NA")
```

## Data filtering

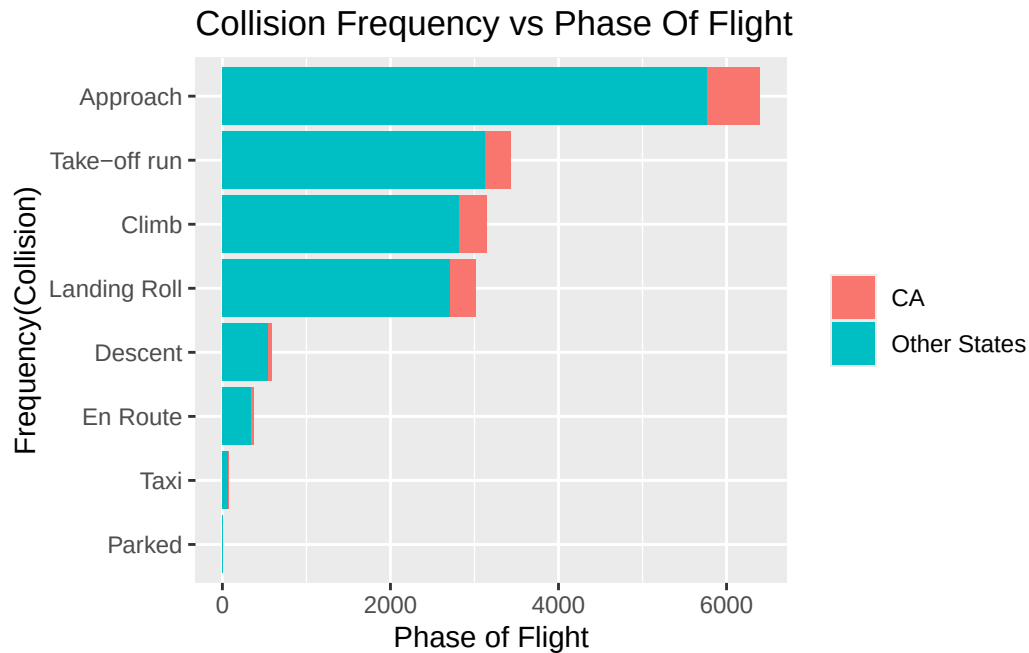
```
plot1_data <- birds |>
  select(phase_of_flight, state) |>
  filter(!is.na(phase_of_flight), !is.na(state)) |>
  mutate(state1 = ifelse(state == "CA", "CA", "Other States"))

plot2_data <- birds |>
  select(phase_of_flight, time_of_day, state) |>
  filter(!is.na(phase_of_flight), !is.na(state), !is.na(time_of_day)) |>
  mutate(state2 = if_else(state == "CA", "CA", "Other States"))
```

## Plot 1

Version 1

```
ggplot(plot1_data,
  aes(x = fct_rev(fct_infreq(phase_of_flight)),
    fill = state1)) +
  geom_bar() +
  coord_flip() +
  labs(title = "Collision Frequency vs Phase Of Flight ",
    x = "Frequency(Collision)",
    y = "Phase of Flight",
    fill = NULL)
```

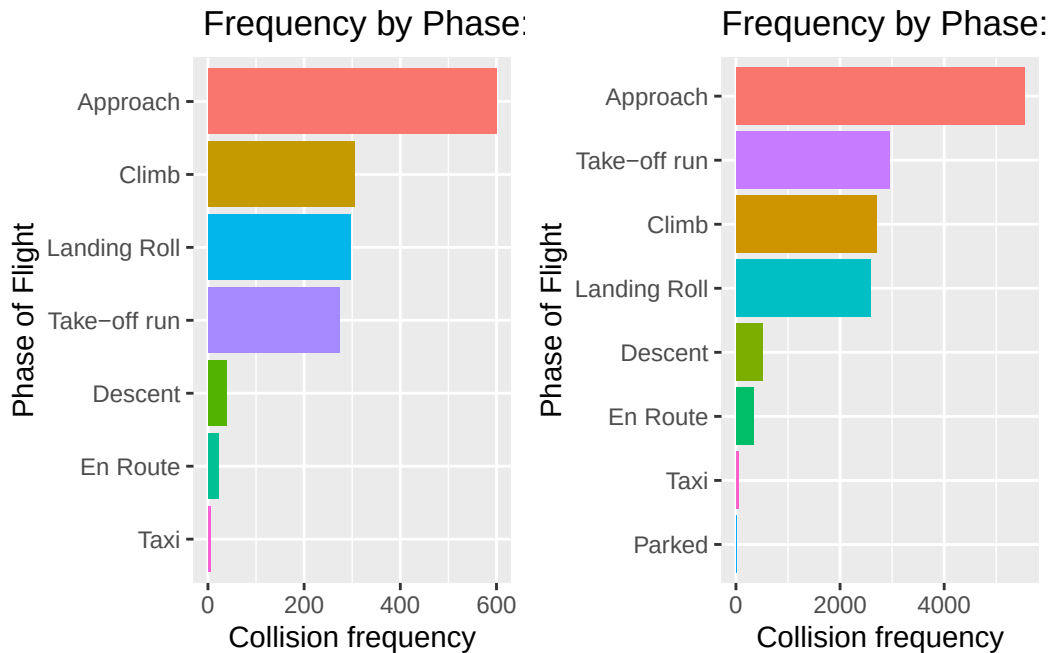


Version 2

```
p1 <- ggplot(birds_CA ,aes(x = fct_rev(fct_infreq(phase_of_flt)), fill = phase_of_flt)) +
  geom_bar(show.legend = FALSE) +
  coord_flip() +
  labs(title = " Frequency by Phase: CA",
       y = "Collision frequency",
       x = "Phase of Flight")

p2 <- ggplot(birds_Otherst ,aes(x = fct_rev(fct_infreq(phase_of_flt)), fill = phase_of_flt)) +
  geom_bar(show.legend = FALSE) +
  coord_flip() +
  labs(title = "Frequency by Phase: Other States",
       y = "Collision frequency",
       x = "Phase of Flight")

grid.arrange(p1, p2, nrow = 1)
```



This plot compares California and other states in terms of which **flight phase** had the most bird collisions.

In both groups, most collisions happen during the **approach** and **landing** phases. This is expected because planes are flying lower and slower then, which increases the chance to hit birds.

In **California**, the collisions are more evenly spread across all phases, while in **other states**, they are mainly in one or two phases, especially during **approach**.

Even if the scales are different, the trend is the same: the **approach phase** has the most collisions in both cases.

## Plot2

```
p3 <- plot2_data |>
  filter(state2 == "CA") |>
  ggplot(aes(x = fct_infreq(phase_of_flt), fill = time_of_day)) +
  geom_bar(position = position_dodge(width = 0.8), width = 0.7) +
  facet_wrap(~state2, ncol = 2) +
  labs(title = "Phase of Flight vs Time of Day (CA)",
       x = "Phase Of Flight",
       y = "Time of Day")
```

```
p4 <- plot2_data |>
  filter(state2 != "CA") |>
  ggplot(aes(x = fct_infreq(phase_of_flt), fill = time_of_day)) +
  geom_bar(position = position_dodge(width = 0.8), width = 0.7) +
  facet_wrap(~state2, ncol = 2) +
  labs(title = "Phase of Flight vs Time of Day (Other States)",
       x = "Phase Of Flight",
       y = "Time of Day")
```

This plot shows how **time of day** affects collisions for California and other states. In both places, most collisions happen during the **daytime**, and there are fewer at **night** or **dawn/dusk**.

In **California**, the collisions during the day are more spread across different flight phases. But in **other states**, they are mostly concentrated in the **approach phase** during the day. So even though the number of collisions is not on the same scale, the pattern is clear: **daytime and approach phase** are the most common times for bird collisions in both California and other states.

```
grid.arrange(p3, p4, nrow = 1)
```

